

LESSON 304 (1.5 AATD)

Lesson Objective

During this lesson, the student will review the maneuvers listed for full and partial panel to increase proficiency. The student will be introduced to additional partial panel maneuvers.

Briefing

- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

Lesson Content (Simulator)

System Power Up

- Student Tells, Student Does.
 - Have the student start the simulator using the checklist and assist.
 - Ensure that the student checks GPS database currency.
 - Half-left, half-up, flag out of view.
 - Ensure that there are no red 'x's.
 - Have the student start the engine using the checklist.

Instrument Cockpit Check

- Student Tells, Student Does.
 - Have the student read each item from the checklist out loud.
 - Demonstrate to the student what needs to be done for each item.
 - Discipline the student to set bugs, bearing pointers, and any auxiliary features.

Before Taxi Checks

- Student Tells, Student Does.
 - Airspeed indicator (PFD and standby): zero.
 - Attitude indicator (PFD): pitch/bank deviations +/- 5 degrees.
 - Standby attitude indicator: discuss standby battery check and adjust the miniature aircraft (cannot be performed in the ATD).
 - Altimeter (PFD and standby): +/- 75 feet of station elevation.
 - Vertical speed indicator (PFD): reads 0 or use current indication.
 - Trim coordinator: wings level or no vector when still; inclinometer centered when still; skids when turning.
 - HSI: matches compass and turns appropriately.
 - Magnetic compass: indicates heading properly, no bubbles, card present, and swings freely.
 - Use a logical flow while checking instruments!
 - Discuss the proper actions to take if a piece of equipment fails its check.

Give the student a departure via radar vectors. The instructor simulates ATC. Have the student switch to the appropriate frequencies throughout the simulator lesson.

Basic Instrument Maneuvers - Full Panel

- Straight and level:
 - Student Tells, Student Does.
 - Remind the student to begin level-off 10% vertical speed.
 - Assist the student with pitch and power settings for the simulator.
 - Allow the student to get a feel for trimming.
 - Training cruise for the appropriate aircraft is to be maintained unless otherwise specified.
 - The appropriate checklists must be completed.
- Compass turns:
 - Instructor Tells, Student Does.
 - Either prior to, in conjunction with, or succeeding the partial panel patterns, compass turns are to be made.
 - All turns should be made using time.
 - The concept of UNOS was introduced in the last lesson. Review as needed.
- Steep turns:
 - Student Does, Instructor Evaluates.
 - Note heading – conduct clearing turns at 30° bank preceding the maneuver.
 - Scan will need to be rapid to remain within parameters.
 - It is best to alternate directions for proficiency.
- Recognition and recovery from unusual attitudes:
 - Student Does, Instructor Evaluates.
 - It is recommended to conduct at least 2, and they may be done at random.
 - This is knowledge from private pilot training, so the student should be able to recover promptly and correctly.

Preceding partial panel training, it is recommended to fail the student's PFD or ADAHRS. Have the student use the Abnormal Checklist.

Basic Instrument Maneuvers - Partial Panel

- Pattern E:
 - Instructor Tells, Student Does.
 - The instructor will call the turns.
 - For this pattern, have the student time turns to headings.
 - Rollout is initiated when time expires.
 - Coach the student on changing the angle of bank when airspeeds change to maintain standard rate.
 - Verify the accuracy of each turn.
 - Upon completion of the maneuver, compare the student's time to the required time.

- Vertical S – Timed Climbs and Descents:
 - Instructor Tells, Student Does.
 - Cover the altimeter – climbs and descents will be conducted with reference to the timer and VSI.
 - The student keeps time.
 - The instructor designates a heading and will call the climbs and descents.
 - After each minute, uncover the altimeter to evaluate the accuracy of the climb/descent.
 - Monitor performance and coach use of pitch and power.
 - Upon completion, note the time and altitude.
- Pattern F:
 - Instructor Tells, Student Does.
 - The instructor will call the turns.
 - The student will track time.
 - This time, the student will use the compass solely to make turns (UNOS) – ensure level-offs are made 10% VSI.
 - Coach the student on changing angles of bank with airspeed changes, and pitch moment changes with flaps.
 - Monitor accuracy of turns and timekeeping.
 - Upon completion of the maneuver, compare the student's time to the required time.
- Vertical S-1:
 - Instructor Tells, Student Does.
 - The instructor will call the climbs, descents, and turns.
 - The student keeps time.
 - Heading and altitude will indicate the accuracy of flying after each minute.
 - It is recommended to use the compass as the main reference for turns in conjunction with the timer.
 - Monitor the flying and coach use of pitch and power.
 - Once complete, note the time.

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 305.
- Assign Homework:
 - Instrument flying handbook:
 - Chapter 07 - Section II.

- Chapter 09.

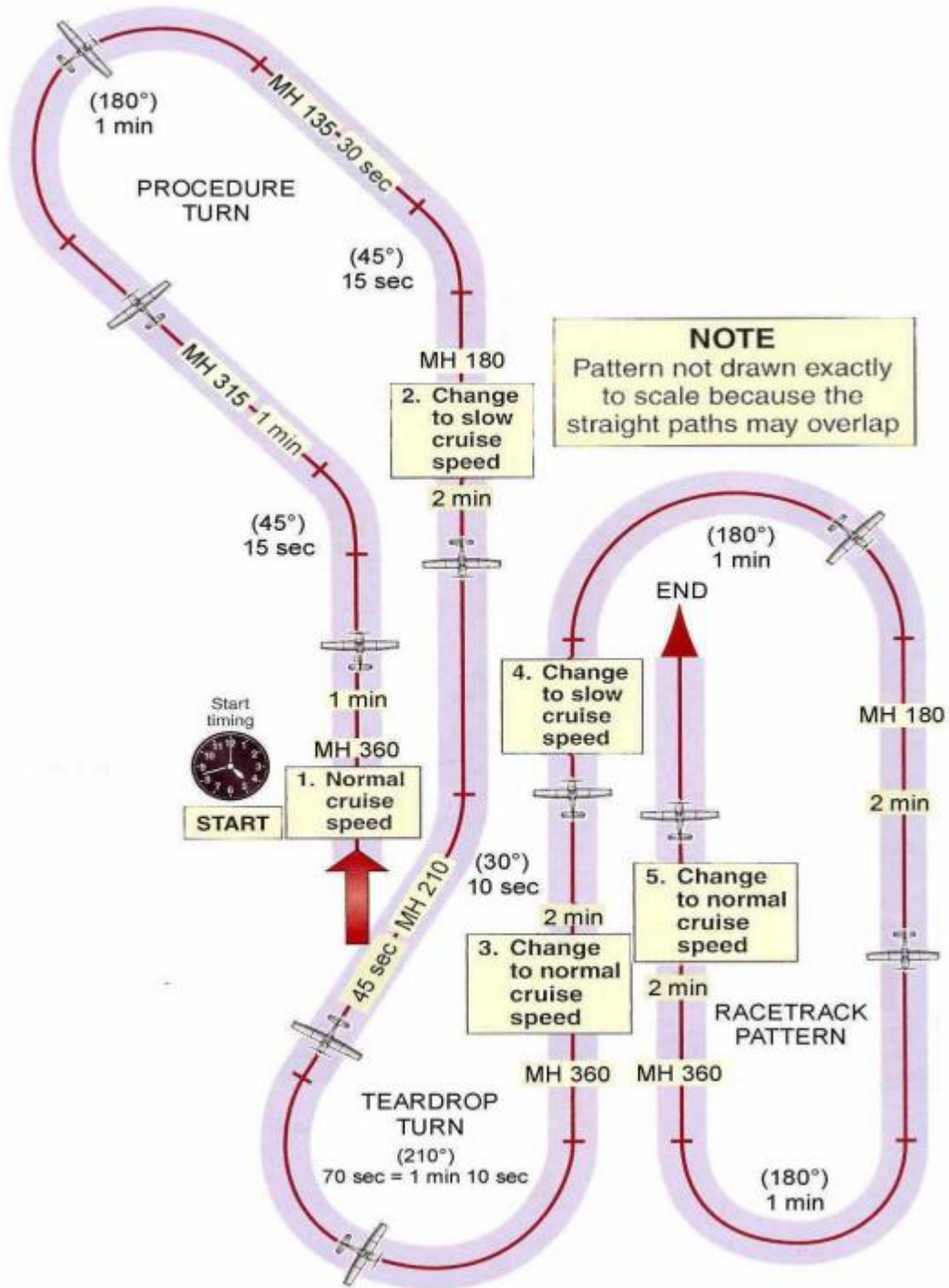
Completion Standards

The student should understand both the change in instrument reference and the interpretation necessary to maintain control while using partial panel.

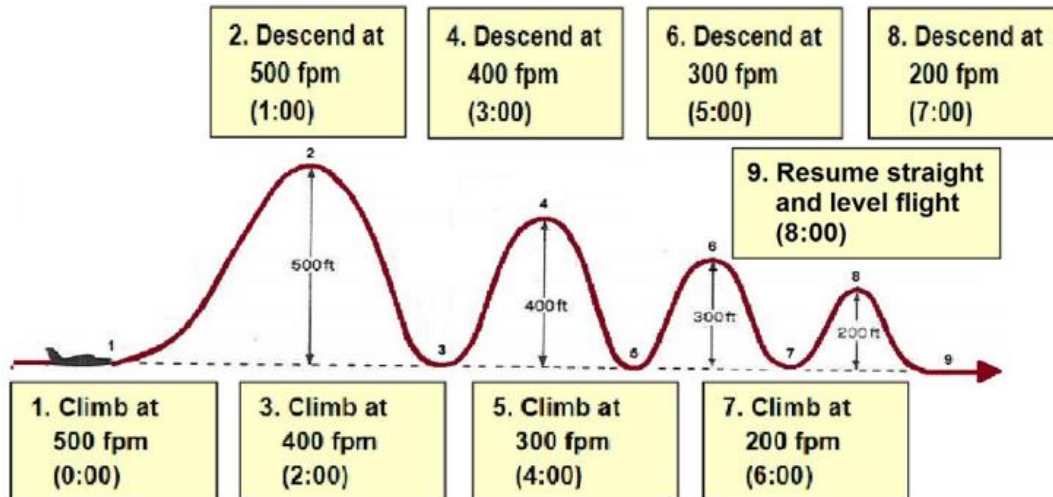
Lesson Plans

Instrument Rating Course

Pattern E



Vertical S



Vertical S-1

